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Developing a jam-absorption strategy for mixed traffic flow at signalized intersections using deep reinforcement learning

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ABSTRACT

Jam-absorption driving (JAD) can effectively prevent the generation and propagation of traffic oscillation. To alleviate the traffic congestion in the signalized intersection with mixed traffic flow, including human driving vehicles (HDVs) and connected and automated vehicles (CAVs), this study provides a jam-absorption driving strategy based on the traffic delay prediction of the mixed platoon under traffic congestion. An online traffic congestion prediction method with the objective of JAD is proposed and focuses on the leaving state of the trajectory to achieve fast capture of congestion features. Then, with real-time status and prediction information, we develop a Jam-absorption driving strategy based on a deep reinforcement learning (DRL) model to improve adaptability to the mixed traffic environment. The results show that this strategy can suppress more than 70% of traffic oscillations with excellent execution efficiency, improving traffic safety and efficiency.

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Jam absorption; traffic oscillation; mixed traffic flow; connected and automated vehicles; signalized intersection; deep reinforcement learning

Introduction

Traffic oscillations refer to the stop-and-go phenomenon where vehicles continuously slow down, stop, and restart. This phenomenon often occurs in unstable traffic conditions, which can lead to severe congestion and negatively impacts traffic efficiency, safety, and energy consumption (Chen et al. 2021; Y. Li et al. 2017; Treiber and Kesting 2011; Xu et al. 2012). Previous studies have primarily focused on alleviating traffic oscillations on freeways by controlling macroscopic traffic states, such as variable speed limit (VSL) control. The main idea of these studies is to smoothly decelerate the upstream traffic flow of traffic oscillations on freeways by dynamically changing speed limits (Han, Chen, and Ahn 2017; Z. Li et al. 2014; Peng and Xu 2022). The major limitation associated with VSL is that it can only change speed limits at fixed and discrete locations, leading to reduced spatial resolution of control input for dissipating traffic flow shockwave. Moreover, drivers' compliance with speed advisory may also reduce the performance of VSL.

With the recent developments in autonomous driving and communication technologies, jam-absorption driving (JAD) for individual vehicles was developed to overcome the limitations of traditional macroscopic traffic state control techniques on freeways (Lu et al. 2020). This strategy was first proposed by Beaty (1998), and its basic idea is that a designated vehicle performs 'slow-in' and 'fast-out' movement within an oscillating region to guide subsequent vehicles' speeds and prevent the propagation of traffic oscillations on freeways. The effectiveness of the JAD strategy was validated in both theoretical (Nishi et al. 2013), Simulation experiment (Z. He et al. 2016) and real-world scenarios (Stern et al. 2018; Taniguchi, Nishi, Tomoeda, et al. 2015). In subsequent studies, researchers have significantly optimized and improved these methods for autonomous driving environments. Several studies have focused on the specification of the key parameters in the JAD process, including the specific vehicle, speed, starting point, and

ending point of JAD execution. For example, Zheng et al. (2020) predicted traffic oscillations based on the Wavelet Transform to determine the starting point, and analyzed the absorption effect of different execution speed under a hypothetical vehicle. However, for a determined traffic congestion, the starting point, ending point and speed are often determined by the selection of vehicle due to the objective of the optimal absorption efficiency. Accordingly, Wang, Li, et al. (2022) proposed an optimization method to find the best absorbing vehicle for JAD with the objective of total delay. The scenarios of these studies are simple single oscillation, while others focus on the extension of JAD strategies to complex traffic scenarios. For secondary oscillation, Taniguchi, Nishi, Ezaki, et al. (2015) analyzed the triggering conditions of secondary oscillations during JAD based on traffic density and vehicle spacing, and established a JAD framework considering traffic flow stability to suppress the occurrence of secondary oscillations. Nishi and Watanabe (2022) proposed a real-time JAD strategy for fixed bottlenecks by constantly updating the optimal selection of absorbing vehicle, but this relies on a high penetration of autonomous driving environments.

While trajectory planning for JAD has effectively suppressed congestion propagation, current researches are limited to continuous traffic flow in expressway environments. In urban intersections, the periodic bottlenecks caused by traffic signals leads to more frequent traffic oscillations compared to freeways. Additionally, dense traffic flow increases the likelihood of secondary oscillations (Mosharafian, Afzali, and Mohammadpour Velni 2023). The aforementioned methods are challenging to directly implement in these environments. Existing researches on trajectory planning at intersections mainly focus on eco-driving. Based on local vehicle information and Signal Phase and Timing (SPaT), optimal trajectory planning is achieved by heuristic algorithms (Lin et al. 2021; Yao and Li 2020) or reinforcement learning (Guo et al. 2021; Wegener et al. 2021). These methods address

optimization problems to enhance energy efficiency in a single vehicle or system while adhering to efficiency and safety constraints, but these studies targeting fuel-reduction or energy-saving have diminished the performance and related discussions of optimal vehicle speed control in suppressing oscillations. Therefore, it is necessary to develop a jam-absorption driving strategy (see Figure 1) for signalized intersection environments to sustainably mitigate the efficiency and safety degradation associated with periodic oscillations.

In mixed traffic environments, human-driven vehicles (HDVs) present a significant challenge to JAD. The queue of uncontrolled HDVs at red lights often causes ‘trajectory jerking’ of CAVs, leading to unnecessary safety and efficiency losses (Wang, Wang, et al. 2022). Previous freeway studies have failed to effectively predict congestion states, and JAD was conducted assuming these states were known. In contrast, traffic congestion caused by SPaT can be predicted over a longer time frame. Therefore, some studies have integrated queuing information into trajectory planning using the Lighthill-Whitham-Richard (LWR) model to estimate queue lengths and mitigate their impacts based on macroscopic traffic flow data (J. Li et al. 2023; Sun et al. 2022; Yang, Rakha, and Ala 2016). However, they only consider the maximum length of the intersection queue, the process of queue dissipation (i.e. vehicles leaving the stop line) is equally important. To this end, a few studies have provided detailed trajectories or departure states of downstream HDVs through car-following model (Wang, Li, et al. 2022) or the combination of shockwave theory and motion prediction (Q. Zhou et al. 2023). In these frameworks, frequent updates and repetitive prediction processes impose a significant computational burden, and macro information struggles to accurately provide micro-level details for trajectory planning. Therefore, how to establish a more efficient method to provide accurate queuing delay information for intersection JAD remains an open problem.

In order to fill the above research gaps, this paper aims to develop an adaptive and real-time JAD strategy for mixed traffic flow at signalized intersections. The proposed JAD strategy can reduce periodic traffic jams caused by signals by deploying multiple jam-absorbing vehicles. An online traffic congestion prediction method was initially developed to forecast vehicle departures from intersections and identify traffic delays using multi-objective trajectory modeling. The predicted leaving state of the preceding platoon provides the CAV with prior information about traffic congestion, which is beneficial to avoid the effects of uncertainty from HDVs. To improve the computational efficiency and adaptability of the strategy, the JAD strategy is developed based on the deep reinforcement learning (DRL) method with multiple objectives of traffic efficiency, safety, and driving comfort. Moreover, the distributed control framework can effectively suppress the periodic

and secondary oscillations by continuously deploying multiple jam-absorption vehicles.

The paper is structured as follows. Section ‘Traffic congestion prediction method’ introduces an online traffic congestion prediction method. Section ‘DRL-based trajectory planning model’ discusses the development and training of the DRL model for jam-absorption driving. Section ‘Simulation and discussion’ presents the experiment design and simulation results, and Section ‘Conclusions’ concludes this study.

Traffic congestion prediction method

This section introduces an online traffic congestion prediction method for signal intersections in mixed traffic. In the mixed traffic environment, unlike HDV, the CAVs not only needs to maintain a safe, comfortable and efficient car-following state, but also needs to achieve the objective of jam-absorption driving. The trajectory planning strategies of CAVs are not accessible for prediction, so car-following models are often used instead to predict complete trajectories. However, it makes the traffic delay prediction more complicated, and the neglect of the optimization objective leads to the accumulation of errors (Anusha, Vanajakshi, and Subramanian 2022). Therefore, a trajectory prediction model with multiple objectives based on Newell’s theory is proposed to generate the prior trajectory state of vehicles leaving the intersection, and a platoon-based traffic delay prediction framework is developed to realize real-time updates.

Vehicle trajectory analysis

In previous studies, the absorption of traffic congestion often adopts the strategy of fast-in and slow-out to inhibit the transmission of traffic shock waves (Wang, Wang, et al. 2022). Under the influence of jam-absorption driving, the spatial-temporal diagram of trajectories at the intersection is illustrated in Figure 2. The key to jam-absorption driving is how to determine the delay time caused by the traffic congestion ahead. The state of vehicle i when it first enters the control area is defined as the initial state (t_A^i, v_A^i) . The states of vehicle i arriving at and leaving the stop line are defined as the arrival state (t_B^i, v_B^i) and the departure state (t_C^i, v_C^i) , respectively. These states can accurately describe the delay of vehicle i at intersection. It should be noted that HDVs in mixed traffic environments do not have the communication capability to obtain their status information in real-time. Therefore, the traffic detector at upstream provides the initial status of vehicles.

Due to the periodic traffic signals, discontinuous vehicle trajectories are divided into approach and departure processes.

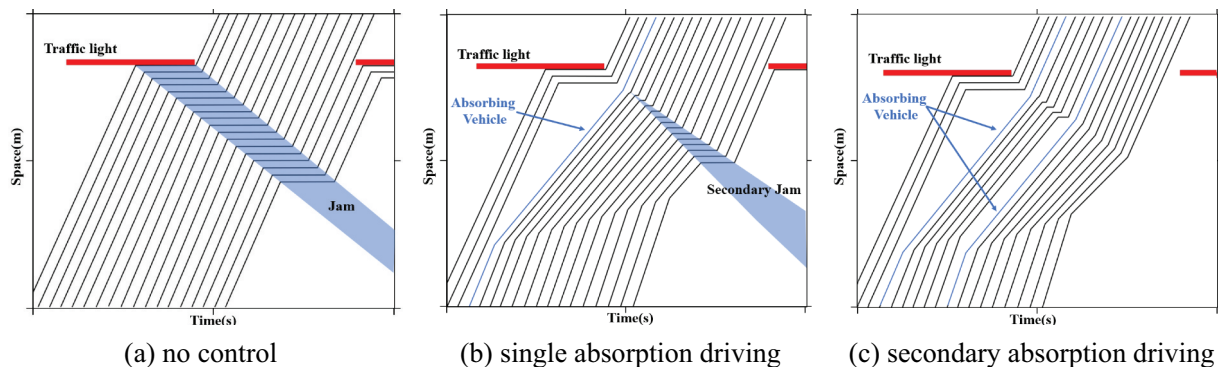


Figure 1. Jam state of intersection under different JAD strategies.

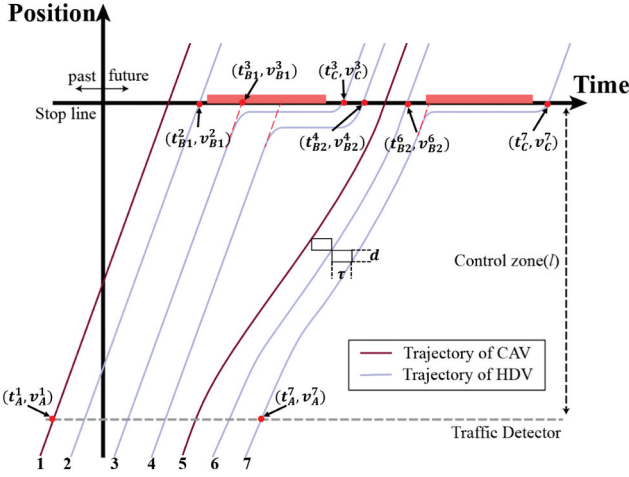


Figure 2. The spatial-temporal diagram of trajectories.

The vehicle trajectory during the approach is characterized by two states: the free state (t_{B1}^i, v_{B1}^i) and the following state (t_{B2}^i, v_{B2}^i) . The free state is the earliest time and corresponding speed to leave the intersection in the free flow based on the initial state (e.g. the status (t_{B1}^2, v_{B1}^2) of vehicle 2 and the status (t_{B1}^3, v_{B1}^3) of vehicle 3 in Figure 2). When the vehicle is in the car-following state, the state of the vehicle arriving at the intersection stop line is affected by the motion of the preceding vehicle (e.g. the status (t_{B2}^4, v_{B2}^4) of vehicle 4 and the status (t_{B2}^6, v_{B2}^6) of vehicle 6 in Figure 2). In the environment of signalized intersection, vehicle trajectories may be affected by the red light. The departure process is also categorized into two states based on whether or not it is affected by red light. When a vehicle's trajectory is not affected by red light, the vehicle leaves the stop line in previous arrived state (i.e. (t_C^i, v_C^i) is the same as (t_B^i, v_B^i)). Some vehicles may need to wait for the end of the red light before leaving the stop line (e.g. the status (t_C^3, v_C^3) of vehicle 3 and the status (t_C^7, v_C^7) of vehicle 7 in Figure 2). Based on the above analysis, the trajectory status satisfies the following relation:

$$t_A^i < t_{B1}^i \leq t_{B2}^i \leq t_C^i \quad (1)$$

Trajectory prediction model

In the past, A large number of car-following models are used to predict the trajectory of vehicles. Newell's car-following theory is the most common theoretical basis for explaining traffic oscillation, which translates the trajectory of the leading vehicle along the direction of time and space to simulate the trajectory of vehicles under congestion (Newell 2002). It is beneficial to predict the end state of the trajectory (i.e. leaving state from the stop line) without relying on the process state.

Based on the analysis in Section 'Vehicle trajectory analysis', the vehicle's arrival state (t_B^i, v_B^i) is selected from the possible free state (t_{B1}^i, v_{B1}^i) and the possible following state (t_{B2}^i, v_{B2}^i) . The arriving state Tr_B^{ω} of a platoon ω with n vehicles is predicted as follows:

$$t_{B1}^i = t_A^i + f(l, v_A^i, v_{\max}, a_{\text{des}}) \quad (2)$$

$$v_{B1}^i = \min\{v_{\max}, v_A^i + a_{\text{des}}(t_{B1}^i - t_A^i)\} \quad (3)$$

$$t_{B2}^i = t_C^{i-1} + \tau + \frac{d}{v_C^{i-1}} \quad (4)$$

$$v_{B2}^i = v_C^{i-1} \quad (5)$$

$$Tr_B^{\omega} = \left\{ \left(\max(t_{B1}^i, t_{B2}^i), v_{B1}^i \cdot I_{[t_{B1}^i \geq t_{B2}^i]} + v_{B2}^i \cdot I_{[t_{B1}^i < t_{B2}^i]} \right) \mid i = 1, 2, \dots, n \right\} \quad (6)$$

Where the function $f(\cdot)$ is the minimum time for the vehicle to travel the target distance with the constraints of speed and acceleration, defined as below. l is the range of control zone (i.e. the distance between the traffic detector and stop line). $v_{\max}$ is the maximum speed. a_{des} is the desired acceleration. τ and d are the constant time gap and space gap of Newell's car-following model. n is the number of vehicles in the platoon ω . The status (t_C^0, v_C^0) is the departure status of the previous platoon $\omega - 1$, which depends on the last vehicle in it. $\mathbb{I}_{[\text{condition}]}$ is an indicator function. When the condition is true, the output is 1; otherwise, the output is 0.

$$f(x, v, v_{\max}, a_{\text{des}}) = \begin{cases} \frac{-v + \sqrt{v^2 + 2a_{\text{des}}x}}{a_{\text{des}}}, & \frac{v_{\max}^2 - v^2}{2a_{\text{des}}} > x \\ \left(\frac{x - \frac{(v_{\max}^2 - v^2)}{2a_{\text{des}}}}{v_{\max}} \right) + \frac{(v_{\max} - v)}{a_{\text{des}}}, & \frac{v_{\max}^2 - v^2}{2a_{\text{des}}} \leq x \end{cases} \quad (7)$$

The motion of vehicles may be affected by red lights in a signalized intersection environment, so the delay caused by red lights needs to be modeled. It is assumed that the signal timing strategy and its current process are accessible. A binary variable $\beta_i^j(t)$ is introduced to describe whether vehicle i in the platoon ω is affected by the delay of red light j . The leaving state (t_C^i, v_C^i) of a platoon ω is:

$$\beta_i^j(t_B^i) = \begin{cases} 1, & t_B^i \in [t_{\text{start}}^j, t_{\text{end}}^j] \\ 0, & \text{otherwise} \end{cases} \quad (8)$$

$$t_C^i = (1 - \beta_i^j) t_B^i + \beta_i^j (t_{\text{end}}^j + t_{\text{rea}} + t_{\text{toline}}) \quad (9)$$

$$v_C^i = (1 - \beta_i^j) v_B^i + \beta_i^j a_{\text{des}} t_{\text{toline}} \quad (10)$$

$$Tr_C^{\omega} = \{(t_C^i, v_C^i) \mid i = 1, 2, \dots, n\} \quad (11)$$

where $t_{\text{start}}^j, t_{\text{end}}^j$ are the start time and end time of the red light j respectively. t_{rea} is the reaction time for vehicle starting. t_{toline} is the time from starting to leaving the stop line for the first car behind the stop line. To simplify the calculation, both t_{rea} and t_{toline} are fixed.

Predictive framework

As mentioned above, the key of JAD is to determine the trajectory state of vehicles leaving the intersection. Therefore, the prediction of the process states of trajectory is ignored. Meanwhile, to improve the prediction efficiency under high traffic volume, only when a CAV enters the control zone, a prediction process is performed to provide it with prior trajectory prediction information. Since the prediction is performed only once and depends on the prediction results of the preceding vehicles, it is easy to cause the accumulation of errors. To this end, a platoon-based prediction pattern was adopted in this framework to limit the accumulation of errors in a platoon. In this study, we designate a platoon as a generic CAV-HDVs pattern to describe mixed traffic flow, which consists of a leader CAV followed by one or multiple HDVs, as illustrated in Figure 3. Alternatively, individual CAVs are also considered as separate platoons when no HDVs follow, exemplified by platoon

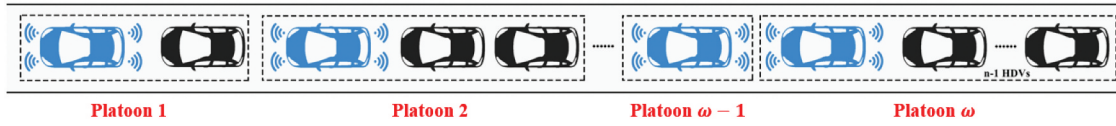


Figure 3. The generic mixed platoon topology.

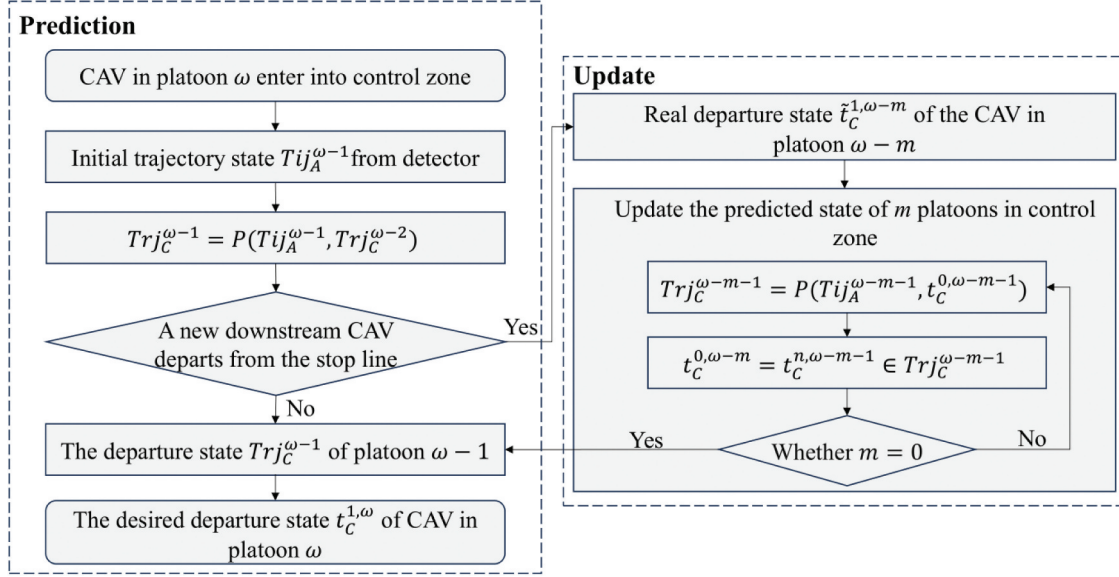


Figure 4. The predictive framework of the mixed platoon.

$\omega - 1$ in Figure 3. Previous definitions of mixed platoon have been limited by vehicle spacing to describe the propagation of traffic oscillations between successive HDVs (M. Li, Cao, and Li 2021; Shi et al. 2023). However, in a mixed traffic environment at intersections, the continuity of the platoon cannot be guaranteed due to the influence of periodic traffic signals. Therefore, the proposed platoon pattern is constructed based on the vehicle distribution information provided by the traffic detectors without being constrained by spatial distance.

The prediction process is shown in Figure 4, including two stages: (1) Prediction. When entering the control zone, the leading CAV in platoon receives the delay information $Trj_C^{\omega-2}$ faced by the previous CAV in platoon $\omega - 1$ and the initial information $Tij_A^{\omega-1}$ of the platoon $\omega - 1$ obtained by the traffic detector. Based on these information, the prediction model $P(\cdot)$ in Section 2.2 will predict the downstream delay state and the current CAV's desired departure state. (2) Update. When a CAV leaves the stop line, the prediction information of all platoons in the control area will be updated based on the real departure status of this CAV. The introduction of real information can effectively limit the accumulation of prediction errors.

DRL-based trajectory planning model

The proposed predictive method in section 2 only determines the starting and ending time of JAD. A trajectory planning model based on DRL method was further developed to optimize CAV trajectory of JAD. This section will discuss the detailed of the DRL scheme and explain how to incorporate traffic congestion prediction in the DRL model.

DRL scheme and formulation

DRL models the actual environment as a Markov decision process, consisting of two interactive objects: the DRL agent (i.e. trajectory planning model) and environment. There are four basic elements for a decision-making process of an agent, including state s , action a , policy π , and reward r .

State and action

The DRL agent learns the environment from the state input s at each time step t , and outputs acceleration of CAV as a control action a . The action is determined by a policy π , which is the probability distribution of actions at the current state. Therefore, it is critical to choose typical features as the unique representation space of the input state, which is conducive to accelerating the convergence of the DRL model.

The state space is represented by several variables reflecting environment state from three aspects in Table 1, including car-following state, traffic light state and predicted departure state. The ego vehicle's speed v_i^t , preceding vehicle's speed v_{i-1}^t and their space headway $d_{i,i-1}^t$ were used to measure car-following state of vehicle i . We select the distance from the ego vehicle to the stop line $d_{i,l}^t$, a signal cycle T_c and the position of current time in a cycle $T_c(t)$ to describe traffic light state. With the state of vehicles, traffic light and traffic detector, the proposed method in section 2 will predict the time when the ego vehicle will leave from stop line. We set the departure time t_{pre}^t as the feature of the predicted traffic congestion state.

Table 1. State features for the DRL model.

State classification	Feature	Description
Car-following state	v_i^t	The speed of vehicle i at time t
	$d_{i,j-1}^t$	The spacing between vehicle i and vehicle $i-1$
Traffic light state	v_{i-1}^t	The speed of the preceding vehicle $i-1$
	$d_{i,light}^t$	The spacing between vehicle i and traffic light
	T_c	Signal cycle of traffic light
Traffic congestion state	$T_c(t)$	The position of time t in a light cycle
	t_{pre}^t	The predicted departure time of vehicle i

Reward function

Typically, the reward function reflects the control objective (Y. He et al. 2024). As described in the previous sections, the effect of the DRL agent is to achieve CAV's jam-absorption driving, while ensuring the safety and efficiency of the driving process. Based on that, the reward r is carefully designed with multiple objectives, including local rewards r_{local} and global rewards r_{global} , i.e. $r = r_{local} + r_{global}$. The global reward r_{global} is a negative reward for vehicle collisions, which can help agents to avoid collisions early.

$$r_{global} = \begin{cases} -10, & d_{i-1,i}^t \leq 0 \\ 0, & \text{otherwise} \end{cases} \quad (12)$$

The global reward can set long-term goals for agents and help avoid local optima during training. However, its sparseness leads to inefficient learning. To address this, denser local rewards are implemented to better describe the movement process of agents. The local reward consists of four aspects, which are speed reward r_{speed} , safe reward r_{safe} , leaving state reward r_{time} , and comfort reward r_{acc} :

$$r_{local} = \alpha_1 r_{speed} + \alpha_2 r_{safe} + \alpha_3 r_{time} + \alpha_4 r_{acc} \quad (13)$$

where $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ are coefficients of the local feature rewards. To ensure the stability of the learning process, we set $\alpha_1, \alpha_2, \alpha_3, \alpha_4 = 0.25$ to ensure $r_{local} \in [-1, 1]$.

The speed reward r_{speed} can encourage the CAV to drive at a higher speed to increase efficiency. the higher speed corresponds to the greater reward value, while both speeding ($> v_{max}$) and slow ($< v_{min}$) driving behaviors are punished. The function can be expressed as:

$$r_{speed} = \begin{cases} \left(\frac{v_i^t}{v_{max}} \right)^2, & v_{min} \leq v_i^t \leq v_{max} \\ -1, & \text{otherwise} \end{cases} \quad (14)$$

where v_{min} is the minimum speed limitation to avoid stopping.

The absorbing velocity is directly determined by the starting point and the ending point of the JAD process, which makes the CAV a safety risk when approaching the front vehicle. Therefore, the multi-stage absorption speed is adopted to reduce the safety risk in the relevant studies (Z. He et al. 2016; Nishi et al. 2013). In this study, we design a safe reward function, which judges the security state by the relationship between actual gap $d_{i-1,i}$ and safe gap d_{safe} , and assigns a reward value:

$$d_{safe}^t = s + v_n^t \times \sigma + \max(0, \frac{(v_n^t)^2 - (v_{n-1}^t)^2}{2a_{max}}) \quad (15)$$

$$r_{safe} = \begin{cases} \frac{d_{safe}^t}{d_{i-1,i}^t}, & d_{i-1,i}^t \geq d_{safe}^t \\ -1, & 0 < d_{i-1,i}^t < d_{safe}^t \end{cases} \quad (16)$$

where s is the standstill safety distance. σ is the system delay to avoid collision caused by the error in braking distance (Moon, Moon, and Yi 2009).

Leaving state reward is the most important part for agent to develop its jam-absorption function, including reward r_p for bypassing congestion and reward r_g for following traffic signal. Reward r_p is designed base on the predictor in section 2, and ensures that CAV arrives at the intersection later than the predicted departure time to suppress the oscillations. Reward r_g about traffic signal is used to encourage the CAV to pass the intersection in the green phase, and to avoid unnecessary stopping. The r_{time} is defined as:

$$t_{arr} = t + f(x_t, v_t, v_{max}, a_{des}) \quad (17)$$

$$r_{time} = \beta_1 r_p + \beta_2 r_g \quad (18)$$

$$r_p = \begin{cases} 1 & t_{pre} \leq t_{arr} \leq t_{pre} + T_c \\ -1 & \text{otherwise} \end{cases} \quad (19)$$

$$r_g = \begin{cases} 1 & t_{g_start} + t_b \leq t_{arr} \leq t_{y_end} - t_b \\ -1 & \text{otherwise} \end{cases} \quad (20)$$

where t_{arr} is the time for a vehicle to arrive at the intersection with following its current state. β_1, β_2 are reward coefficients of 0.4 and 0.6 respectively. t_{pre} is the predicted departure time calculated by the predictor. t_{g_start}, t_{y_end} are the start time of green phase and the ending time of yellow phase respectively. In this paper, we allow the CAVs to pass through the intersection at the yellow phase; t_b is a buffer time, which is used to avoid errors, and its value is set to 1s.

The comfort reward makes driving smoother. Because the change rate of speed (i.e. acceleration) has a great influence on the comfort of the passengers (Zhu et al. 2020), we use acceleration to computed the comfort reward as:

$$r_{acc} = -(a_i^t / a_{max})^2 \quad (21)$$

where a_i^t is the acceleration of vehicle i at time t , i.e. the action of agent; a_{max} is maximum acceleration limit.

Development of DRL model

The DRL model continuously optimizes the action policy to maximize the agent's rewards. Considering the requirement for better comprehensive performance and the requirement of the action space (continuous/discrete), this paper adopts the Proximal Policy Optimization (PPO) algorithm to update the action policy. The PPO algorithm is one of DRL algorithm using Actor-Critic architecture (Schulman et al. 2017), which is consist of an actor network with parameter θ and a critic network with parameter ϕ . The actor network updates the policy with objective $L_a(\theta)$ and the critic network is use to evaluate the policy with objective $L_c(\phi)$.

$$L_a(\theta) = \hat{E}_t [\min(\rho_t(\theta) \hat{A}_t(s_t, a_t), \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t(s_t, a_t))] \quad (22)$$

$$\hat{A}(s_t, a_t) = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t) \quad (23)$$

$$L_c(\phi) = \hat{E}_t [r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t)]^2 \quad (24)$$

where $\rho_t(\theta) = \pi_\theta(a_t|s_t) / \pi_{\theta_{old}}(a_t|s_t)$ is the probability ratio of the current policy $\pi_\theta(a|s)$ and old policy $\pi_{\theta_{old}}(a_t|s_t)$. The function $\text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)$ minimize the gap between the old policy and current policy by clipping the probability ratio with the clipping parameter ϵ , reducing the influence of large gradient on update stability. $\hat{A}_t$ is the advantage function of policy at state s_t with the

estimated value $V_\phi(s_t)$ from the critic network. r_t is the instant reward in timestep t . γ is the discount factor.

In PPO, we adopt stochastic strategy of action, i.e. the action is randomly selected from the probability distribution. Although the model reaches a convergences state in training, it is still possible for agent to output unsafe actions that lead to collisions. On the one hand, this safety risk is unacceptable for CAV, and on the other hand, the collision makes simulation reset and affects the training efficiency (Zhu et al. 2020). Therefore, we set up safety check base on Eq (15) and implement mandatory corrections for actions with potential risk. The final action a_i^t is define as:

$$a_i^t = \begin{cases} a_i^t, & d_{i-1,i}^t \geq d_{\text{safe}}^t \\ -a_{\text{max}}, & d_{i-1,i}^t < d_{\text{safe}}^t \end{cases} \quad (25)$$

We construct the urban signalized intersection simulation environment in SUMO. SUMO can provide numerous car-following and lane-changing models, interacts well with other software to enable secondary development. Therefore, it is widely used in various existing studies (M. Li, Cao, and Li 2021; Peng and Xu 2022). The train and test experiments of the proposed model based on PPO algorithm are implemted by Python 3.7 on a computer with 16 GB of RAM and an NVIDIA 3060 GPU. In order to improve the stability of DRL model, we increased the randomness of the traffic light and traffic flow during training. For traffic light, red and green phase can be randomly initialized within the range of 30–60 seconds in each iteration, and yellow phase is always 5 seconds. The number of vehicles loaded into the simulation per second obeys the Poisson distribution with parameter λ :

$$P(x = k) = \frac{e^{-\lambda} \lambda^k}{k!} (k = 0, 1, 2, \dots) \quad (26)$$

where parameter λ can be randomly initialized within the range of 0.10 to 0.25 in each iteration.

The DRL model adopts a distributed framework for learning, which updates the policy of multiple parallel agents. Due to the homogeneous CAVs, all parameters are shared between different agents. The overall learning framework is presented in Figure 5. Agents acquire complete state information in the simulation environment and the predictive model, and take actions to perform jam absorption driving. After implementing actions, agents acquire the

corresponding reward and next state information. The replay buffer receives transfer trajectories ($\{s_i^t, a_i^t, s_i^{t+1}, r_i^t\}$) from mutple parallel agents at each step, which is sampled for updating agent.

The setting of hyperparameters in training process is shown in Table 2. Due to the sparsity of collision penalty, the discount factor is chosen based on the impact step of the collision penalty according in the Bellman equation. The clip parameter of 0.2 refers to the sensitivity analysis experiment about this parameter for PPO in study of Schulman et al. (2017). The policy ratio is restricted to [0.8, 1.2] to avoid too large variation in the policy to affect the stability of training. Similarly, in order to ensure the efficiency and stability of the training, the other hyperparameters were adjusted relying on empirical and values commonly used in studies about DRL (Mousa et al. 2020; M. Zhou, Yu, and Qu 2019).

Simulation and discussion

In this section, the proposed JAD method is discussed in detail from three aspects: (i) the accuracy of departure time prediction method; (ii) the training result of DRL model based on the simulation; (iii) model performance analysis. We comprehensively analyzed the effectiveness and generalization of the proposed method, and compared it with the strategy in the previous study. The comparison strategies include:

- Baseline is the no control scenario where all vehicles are HDV.
- Strategy I is the simplified method in this study without the traffic delay prediction.
- Strategy II is the DRL method based on complete trajectory prediction. Specifically, it takes all vehicles between the target CAV and intersection as an HDV platoon, and simulates the complete trajectories of the platoon to identify the traffic congestion.
- Strategy III is the proposed method.

The simulation experiments are conducted in an isolated intersection with a long approach which can provide sufficient absorption distance for JAD. The shorter absorption distance can lead to the lower absorption speed and affects the suppression effect of traffic congestion (Zheng et al. 2020). We set up a one-lane road with a length of 1600 m and the traffic light is

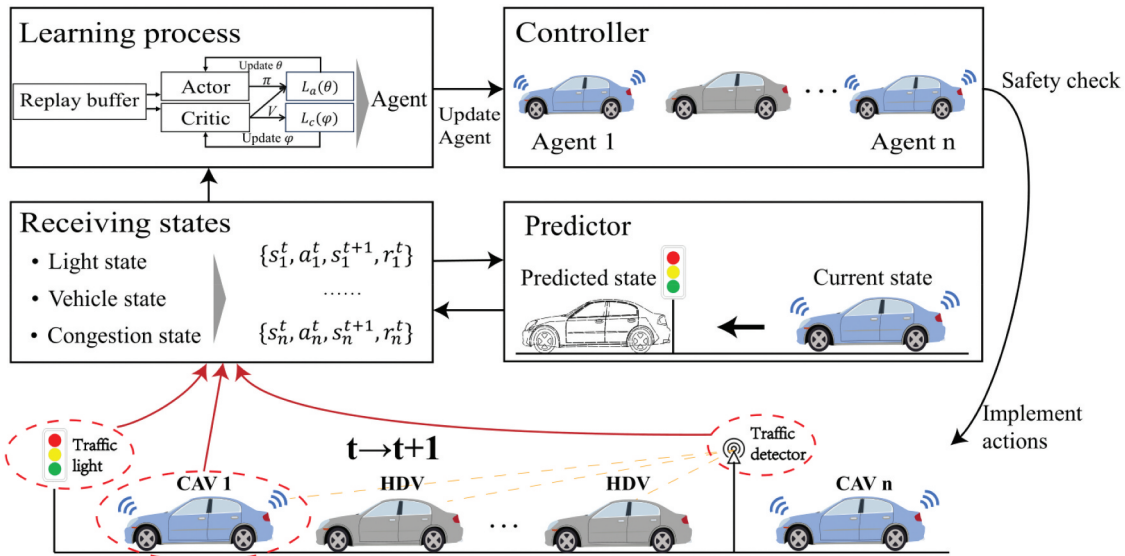


Figure 5. The schematic diagram of training framework.

Table 2. Hyperparameters for PPO.

Hyperparameters	Value
Actor learning rate/ ϵ_a	0.0001
Critic learning rate/ ϵ_c	0.0001
Discount factor	0.97
Clip parameter ϵ	0.2
Replay buffer size	20000
Minibatch size	2000
Hidden layer of actor	2
Hidden layer of critic	2
Number of neurons in hidden layers	128

Table 3. Parameters setting for the experiment.

Parameters	Description	Value
Δt	Step length	0.5 s
$v_{\max}$	The maximum speed limit	22.22 m/s
l	The range of control zone	1000 m
a_{des}	The desired acceleration	1.5 m/s ²
d	Constant space gap	6.6 m
τ	Constant time gap	1.5 s
t_{rea}	Reaction time for vehicle starting	0.5 s
t_{toline}	The time from starting to leaving the stop line	1.5 s
σ	The system delay	0.1 s
s	The standstill safety distance	2.5 m
$v_{\min}$	The desired minimum speed of CAV	2.0 m/s
$[a_{\min}, a_{\max}]$	Acceleration range	$[-3.0 \text{ m/s}^2, 3.0 \text{ m/s}^2]$

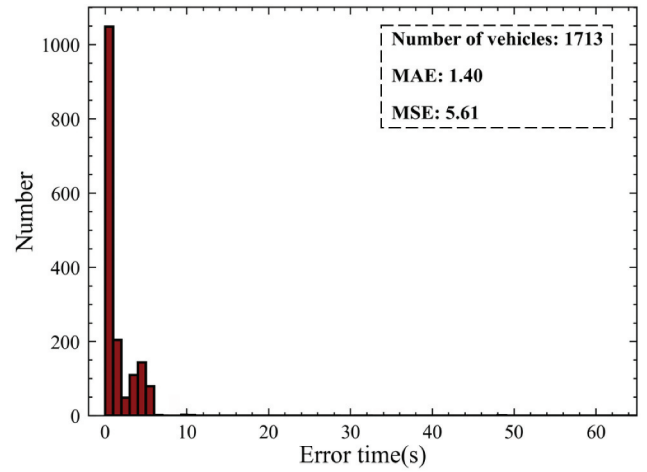
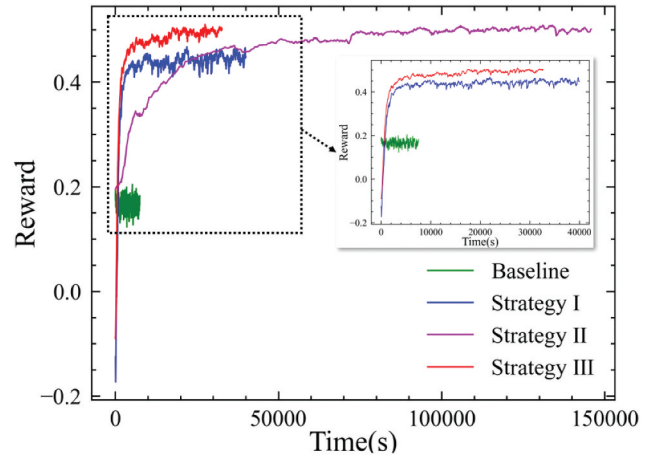
located at 1500 m. The CAV executes the JAD strategy from the traffic detector which is set at 1000 m away from the intersection. The CAV is randomly generated in the traffic flow, and its number is determined by the penetration rate. The parameters setting is shown in Table 3.

Prediction results

Before the training, we verified the accuracy of the proposed prediction method. In the 20 random experiments, a total of 1713 CAVs were counted for leaving time errors from the stop line. The distribution of prediction errors is shown in Figure 6. The error time is concentrated in the range 0–5 seconds, and its average value is 1.40s. In general, the prediction accuracy of this method is high. At the same time, by adjusting the weight of prediction time in the model and introducing the real information to reverse update, the influence of errors in the model training process is further reduced.

Training results

Based on the above training framework and environment, we trained the proposed PPO model and two models for comparison. In the training, the max steps for each episode is 1000s, if a crash occurs, the ego train episode is terminated. The normalized reward curve of 1000 episodes in Figure 7 describes the training process. Figure 7 illustrates that the reward for adopting the control strategy is significantly higher than that of the Baseline, indicating that the implementation of JAD can effectively enhance the efficiency and safety of signalized intersections. Furthermore, when compared to Strategy I, the proposed strategy demonstrates an improved effectiveness in trajectory planning by incorporating delay prediction. In terms of computational efficiency, the reward curves for Strategies I and III stabilize after approximately 10,000 seconds, whereas

**Figure 6.** Distributions of error time.**Figure 7.** Reward curves.

Strategy II converges after about 100,000 seconds. Notably, strategy III completes 1,000 episodes of training at the fastest speed. These results indicate that the proposed strategy significantly outperforms the other compared strategies in terms of computational speed and optimization performance.

Model performance

This section verifies the efficiency and generalization of the proposed model in different environments and analyzes comparison with other control strategies. In this paper, the performance of the model in terms of congestion absorption is evaluated by the stop time and the number of vehicles passing through the intersection. The stop time is the sum of the time that vehicle is in the parking state in the simulation experiment, which reflects the total duration of the stop and go wave in the simulation. In this study, a vehicle is considered to be in a parking state when its speed is below 2 m/s. Efficiency, safety, and comfort performance of vehicle control are discussed using average speed, the time exposed time-to-collision (TET) and acceleration distributions, respectively. TET is the sum of the seconds for which the time-to-collision (TTC) in the simulation is below the safety threshold (Minderhoud and Bovy 2001), and a lower value indicates a safer situation. In this study, the safety threshold is set to be 3s (Autey, Sayed, and Zaki 2012).

Based on default environment parameters, the experiment shows that when the traffic volume reaches 900 veh/h, traffic flow in the

intersection is saturated and continuous congestion is formed, as shown in Figure 8(a). Severe supersaturation does not provide additional absorbing space for the absorbing vehicle, in which case JAD is ineffective (Chen et al. 2021). Thus, the traffic volume of simulation for congestion absorption is set to 900 veh/h in this paper.

Figure 8 shows the absorption effects of different strategies with 20% penetration. The proposed strategy (strategy III) executes JAD at 1000 m away from the intersection (i.e. the detector position). The CAV can pass the signalized intersection without stopping by slowing down in advance, and induce the speed of the following HDVs. Compared with the scenario without control, the congestion at the intersection is greatly reduced in both strategy II and strategy III scenarios. However, due to the limited number of CAVs at low penetration, oscillations cannot be completely suppressed. As shown in Figure 8(b), since CAVs in scenario with strategy I cannot directly obtain congestion information, the starting time of the absorption driving is too late. The JAD in strategy II and strategy III start at 1000 m away from intersection, while strategy I usually starts at a closer position to the intersection. Shorter absorbing distance means lower absorbing speed, and the lower absorbing speed may cause adverse effects on subsequent vehicles (Zheng et al. 2020). In the following, we discuss this phenomenon through specific indicators.

The market penetration rate is the crucial element for mixed transportation system. In the low penetration scenarios, a large number of HDVs is a challenge for the control model. On the contrary, in the high penetration scenarios, CAV can obtain more environmental information base on the V2V technology, which makes the model performance better, as shown in Figure 9. Therefore, we analyzed the performance of the proposed model in different penetrations.

We calculated the average measurements of 20 simulation experiments with a simulation time of 1000s and a traffic demand of 900 veh/h. For the performance of jam absorption, we can see from the Figure 10(a) that the absorbing vehicle can greatly reduce the duration of the stop and go wave in the intersection. Especially at low penetration, strategies II and III can still reduce the stop time by about 74% compared to the uncontrolled scenario, which is better than the 60% improvement in the strategy I scenario. It

means that the provision of jam information is very helpful to improve the model's ability to absorb jam.

The most important benefit of jam-absorption is the improvement of traffic efficiency. Due to the decrease of the stop phenomenon, all strategies have a significant improvement in traffic volume compared to baseline, as shown in Figure 10(b). The volume increases with the increase of the penetration rate and tends to be saturated. Strategies II and III are consistent in terms of efficiency, and the maximum traffic volume can reach 820 veh/h, it has increased by 8% to 10% compared to baseline. Ideally, in an environment with little or no HDVs, strategy I should perform as same as the other two strategies. However, from the results, the efficiency of strategy I is lower than that of the other two strategies in each penetration. We consider that the lack of congestion information in the process of model training may lead to the insufficient adaptability of the model. The performance of the speed is essentially the same as it of the volume as shown in Figure 10(c).

Compared to strategy II, we further focus on the safety performance of the model during the development and validation process. Figure 10(d) shows that the proposed strategy significantly improves the safety performance of the intersection. At 20% penetration rate, the TET of the proposed strategy has a 55% reduction compared with the strategy II. Because the same safety reward function is used in strategy I and strategy III to optimize the safety performance, strategy III still has excellent safety performance, even though it has more traffic oscillations at low penetration.

The improvement in the driving comfort is described by the distribution of vehicle deceleration in the experiment. Generally, a lower deceleration action with longer duration is more comfortable than a shorter with a higher amplitude. As shown in Figure 11, the frequency of high deceleration ($< -1 \text{ m/s}^2$) in the proposed strategy is reduced by 74% compared to the frequency in the baseline strategy. Therefore, the proposed strategy provides a comfortable driving process. The increase in the frequency of deceleration equal to 3 m/s in the figure is due to the forced modification of the safety detection algorithm for potential risky actions in the simulation experiment as in the training process. Comfort performance is less important when the vehicle is at risk.

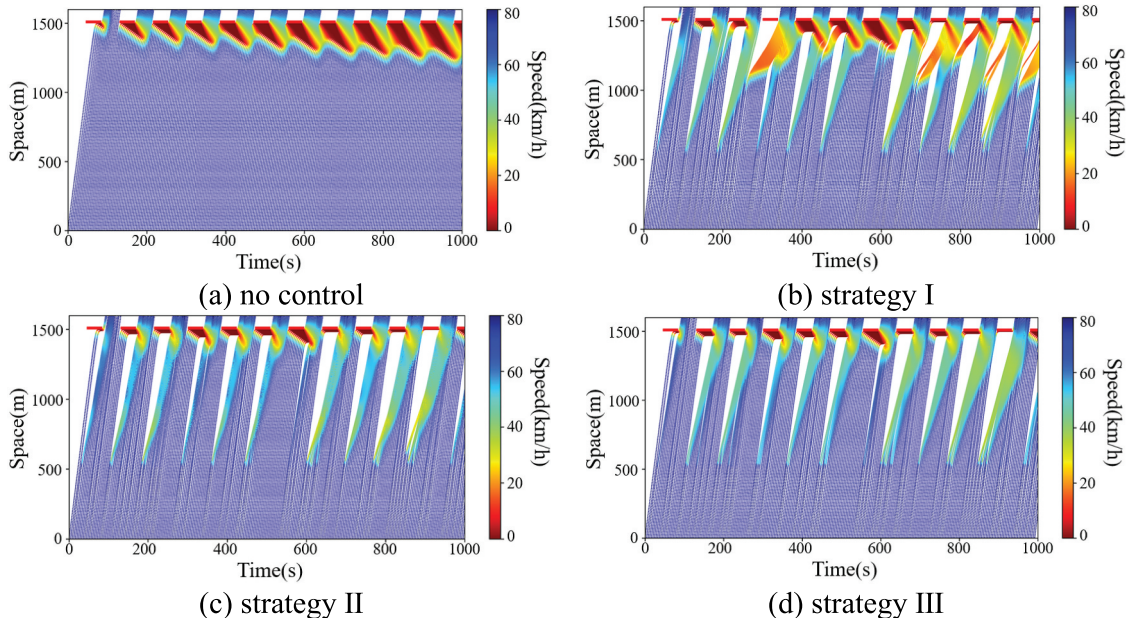


Figure 8. Trajectories of different strategies with 20% penetration.

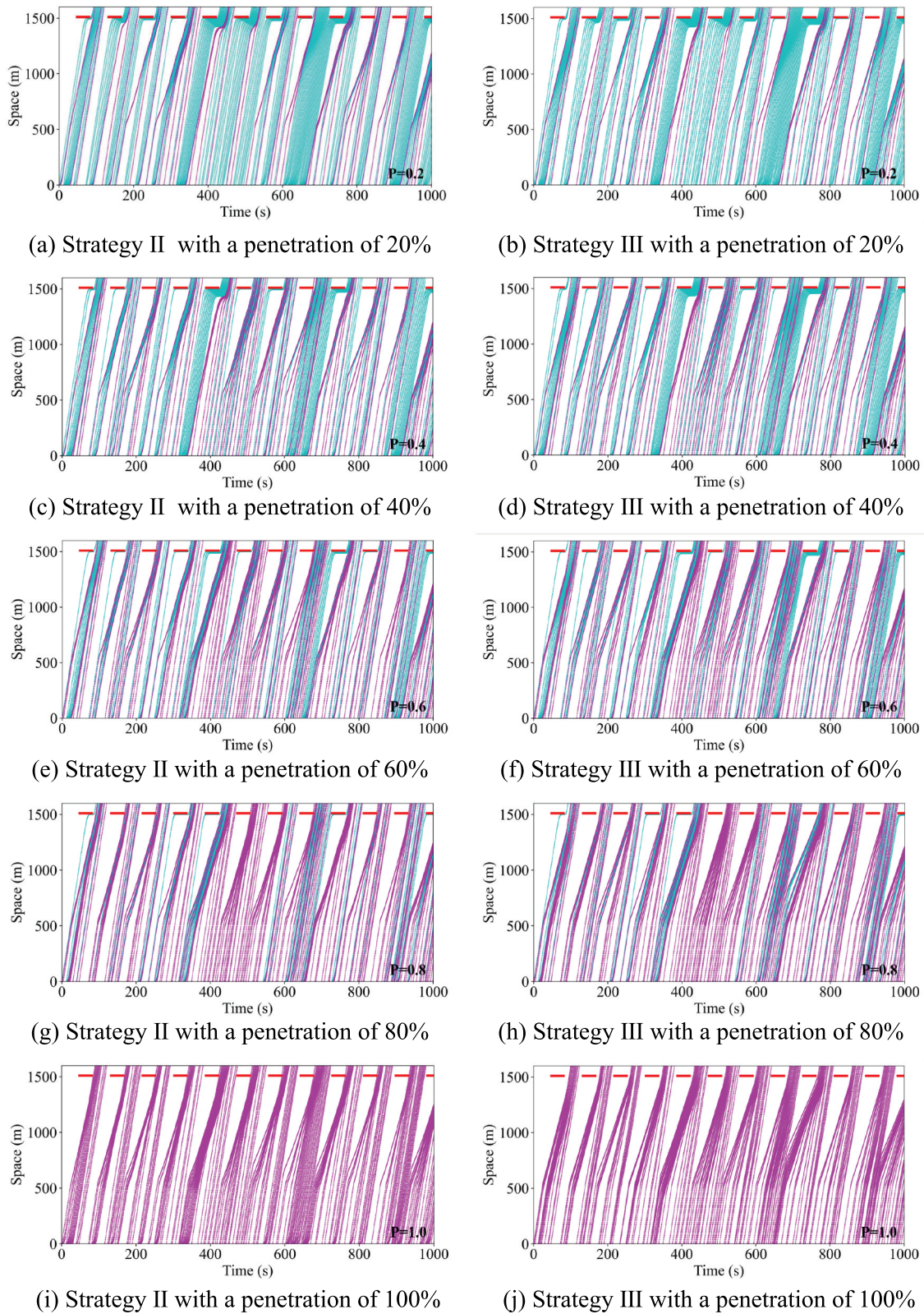
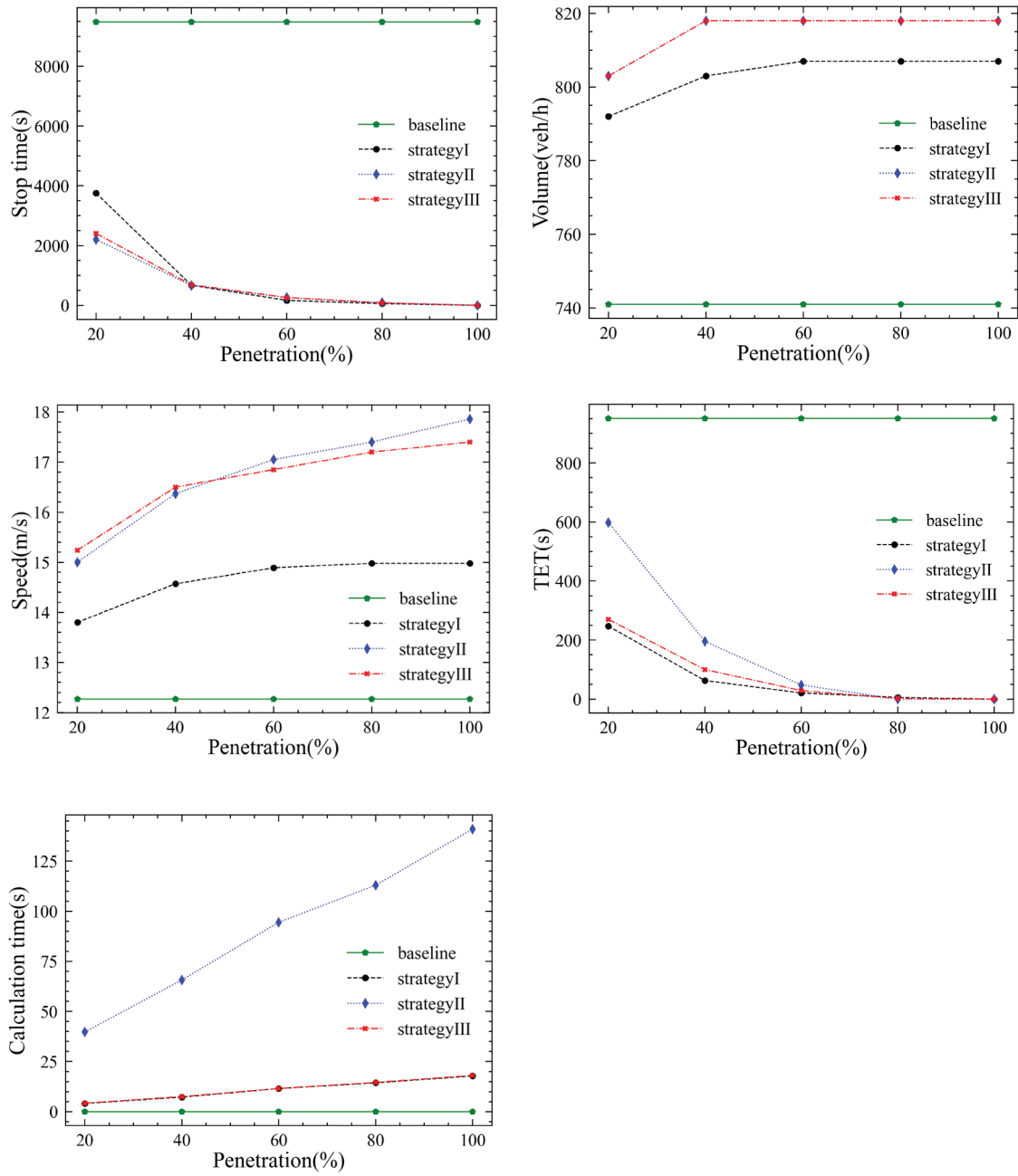


Figure 9. Trajectory in random traffic flow environment with parameter $\lambda = 0.2$.



(e) the performance of the calculation time

Figure 10. The performance comparison of the different strategies at difference penetrations.

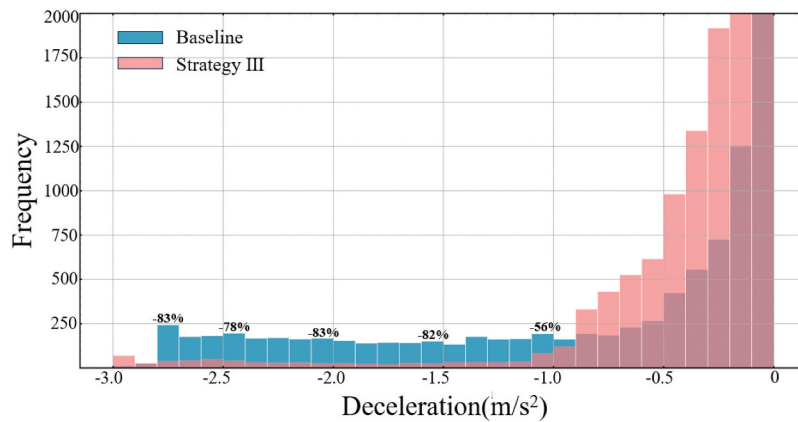


Figure 11. Comparisons of the frequency of different decelerations.

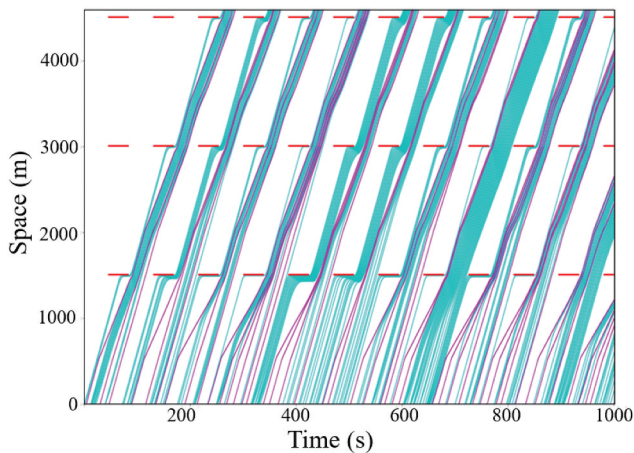


Figure 12. Trajectory of the proposed method in the multi-intersection scenario.

In fact, most intersections are in a continuous intersection environment. Therefore, a three-intersection environment is designed to further verify the generalization ability of the proposed strategy. As can be seen from the vehicle trajectories in Figure 12, the model can still stably implement the jam absorption in this environment.

Finally, Considering the real-time control for CAV, we test the performance of the model in computation time. We show the average calculation time of a simulation experiment in Figure 10 (e). Because we use a simplified trajectory prediction method that only predicts departure time instead of IDM-based full trajectory prediction, the calculation speed of the proposed strategy is greatly improved compared with strategy II. The average time for a control step of strategy is 0.0012s, and it is enough for the online application of the proposed strategy.

Conclusions

This study proposes a jam-absorption framework to suppress the traffic congestion of mixed traffic at a single signal intersection. This framework consists of a traffic congestion predictor and a DRL trajectory planning model, in which predictor can provide the planning model with leaving trajectory state information of the forward platoon for congestion recognition, and then the DRL model plans a safe and stable jam-absorption trajectory for CAVs.

The proposed model is evaluated through simulation experiments in terms of three aspects: jam-absorption, trajectory performance of JAD, and running efficiency of the algorithm. The simulation experiment results show that the proposed model can effectively suppress congestion and improve the traffic capacity of the intersection. At the same time, it can optimize the safety, efficiency and driving comfort of the trajectory. Compared to the existing methods, the proposed model also has a significant improvement in computational efficiency due to the improved method of HDV information acquisition and embedding, and it is more suitable for practical engineering applications.

However, the study still has shortcomings. On the one hand, most of experimental parameters are based on simulation studies. It is necessary to verify the performance of the model through real traffic data. On the other hand, the proposed method is not suitable for heavily congested traffic conditions. Only relying on JAD strategy cannot solve the serious traffic congestion problem, it is

necessary to explore a JAD strategy combined with other control measures (such as traffic signal optimization).

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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